

Orion Main

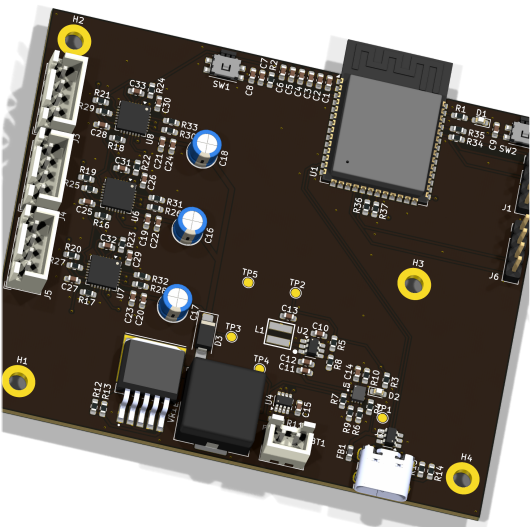
Variant:

2026-09-20

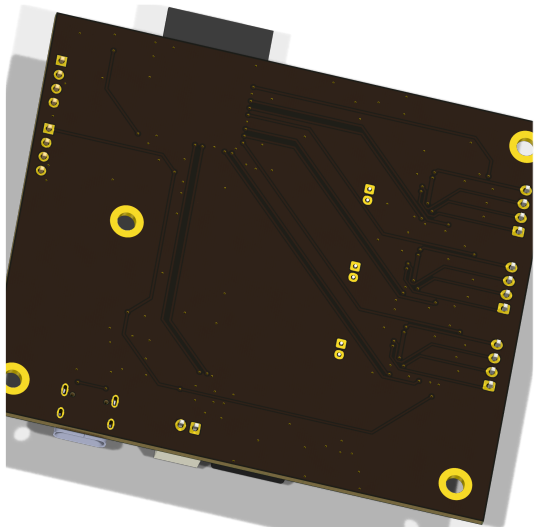
Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 20-Sep-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational design notes.

DESIGN NOTE:
Example text for debug notes.

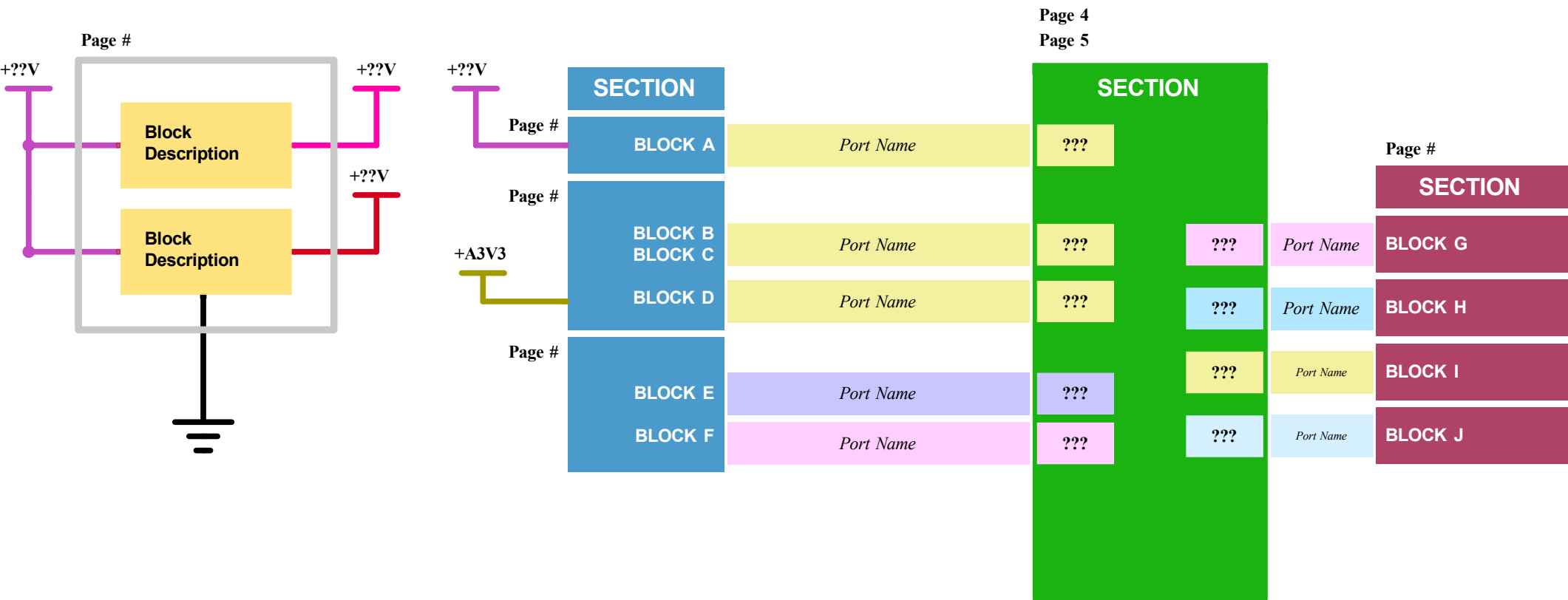
DESIGN NOTE:
Example text for cautionary design notes.

DESIGN NOTE:
Example text for critical design notes.

LAYOUT NOTE:
Example text for critical layout guidelines.

	Comments:	Company: Karpát Space Program		Variant:	Git Hash: e462fed
	Sheet Title:	Board Name: Orion Main		Project Name: Startracker Orion	
	Sheet Path: /	File Name: Orion.kicad_sch	Designer: Bertalan Nagy	Date: 2025-01-12	Revision: + (Unreleased)
			Reviewer:	Size: A3	Sheet: 1 of 7

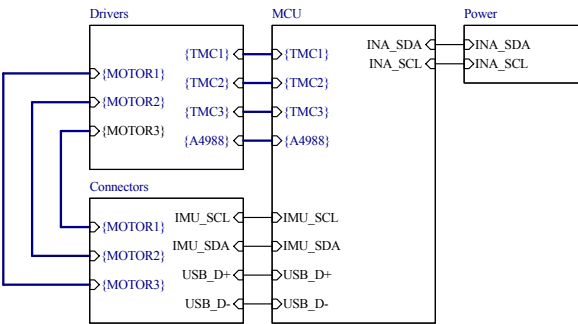
[2] Block Diagram



Target specifications:	
Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

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	Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Bertalan Nagy	Date: 2025-01-12	Revision: + (Unreleased)
Sheet Path: /Block Diagram/			Reviewer:	Size: A3	Sheet: 2 of 7

[3] Project Architecture



	Comments:	Company: Karpát Space Program		Variant:	Git Hash: e462fed
		Board Name: Orion Main		Project Name: Startracker Orion	
	Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Bertalan Nagy	Date: 2025-01-12	Revision: + (Unreleased)
	Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 7

[4] MCU

The schematic illustrates the MCU section of the Orion Main board. Key components and connections include:

- Microcontroller (U1):** ESP32-S3-WROOM-1, with pins 1 (GND), 2 (3V3), 3 (EN), 4 (IO0), 5 (IO1), 6 (IO2), 7 (IO3), 8 (IO4), 9 (IO5), 10 (IO6), 11 (IO7), 12 (IO8), 13 (IO9), 14 (IO10), 15 (IO11), 16 (IO12), 17 (IO13), 18 (IO14), 19 (IO15), 20 (IO16), 21 (IO17), 22 (IO18), 23 (IO19), 24 (IO20), 25 (IO21), 26 (IO22), 27 (IO23), 28 (IO24), 29 (IO25), 30 (IO26), 31 (IO27), 32 (IO28), 33 (IO29), 34 (IO30), 35 (IO31), 36 (IO32), 37 (IO33), 38 (IO34), 39 (IO35), 40 (IO36), 41 (IO37), 42 (IO38), 43 (IO39), 44 (IO40), 45 (IO41), 46 (IO42), 47 (IO43), 48 (IO44), 49 (IO45), 50 (IO46), 51 (IO47), 52 (IO48).
- Power Regulation:** A 3.3V regulator with a 100nF capacitor (C8) and a 100nF capacitor (C9) connected to the EN pin. The regulator is powered by a 3.3V source and has a 100nF capacitor (C7) connected to its output. The output is connected to the 3V3 pin of the MCU.
- USB Interface:** The MCU's USB pins (IO17, IO18, IO19, IO20, IO21, IO22, IO23, IO24, IO25, IO26, IO27, IO28, IO29, IO30, IO31, IO32, IO33, IO34, IO35, IO36, IO37, IO38, IO39, IO40, IO41, IO42, IO43, IO44, IO45, IO46, IO47, IO48) are connected to a USB interface. The interface includes a 47k resistor (R37) and a 47k resistor (R36) connected to the 3.3V source.
- Boot Section:** A boot switch (SW2) is connected to the 3V3 pin of the MCU. A 100nF capacitor (C9) is connected to the SW2 pin.
- Debug and Control:** A debug LED (DEBUG_LED1) is connected to the 3V3 pin of the MCU. A reset switch (SW1) is connected to the EN pin of the regulator.
- Stepper Motor Control:** The MCU's pins are connected to three stepper motors (STEP1, STEP2, STEP3) and their direction pins (DIR1, DIR2, DIR3). The pins are connected to the STEP1, STEP2, STEP3, DIR1, DIR2, DIR3, TX1, TX2, TX3, RX1, RX2, RX3 pins of the stepper motors.

Comments: Sheet Title: Sheet Path: /Project Architecture/MCU/	Company: Board Name: Orion Main		Variant: Project Name: Startracker Orion	Git Hash: e462fed
	File Name: mcu.kicad_sch	Designer: Bertalan Nagy	Date: 	Revision:
	Reviewer: 		Size: A4	Sheet: 4 of 7

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				Board Name: Orion Main			Project Name: Startracker Orion	
				Sheet Title:	File Name: mcu.kicad_sch	Designer: Bertalan Nagy	Date:	Revision:
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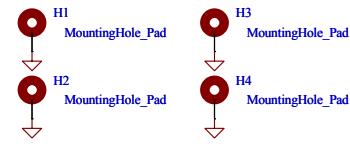
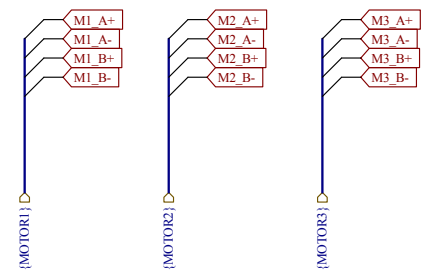
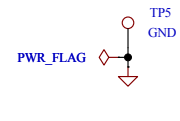
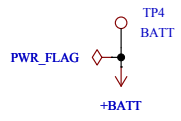
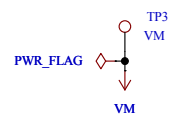
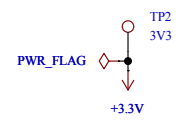
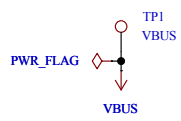
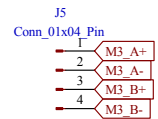
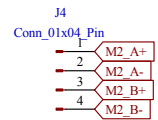
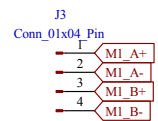
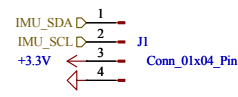
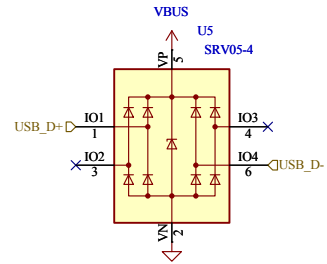
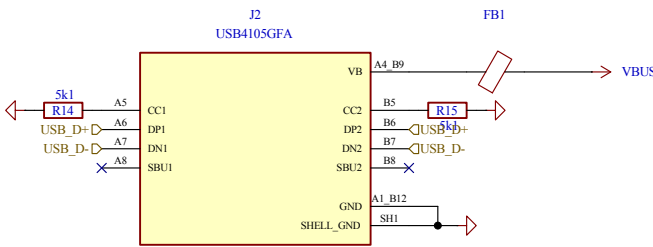
[5] Power

The schematic illustrates the power management system for the Orion Main board. It features three main power sources: VBUS, +BATT, and a battery (BT1). The VBUS input is regulated by the TPS2121RUXR (U3) to provide a +3.3V output. The +BATT input is regulated by the LMR51420 (U2) to provide a +3.3V output. The battery (BT1) is connected to the INA219BxDCN (U4) current sensor, which is also connected to the +BATT line. The output of the current sensor is connected to the VM output. The schematic includes various passive components such as resistors (R4, R6, R7, R9, R10, R12, R13), capacitors (C1, C2, C10, C11, C12, C13, C14, C15), and an inductor (L1). A table at the bottom right contains project metadata.

Comments:	Company:	Variant:	Git Hash:
			e462fed
Sheet Title:	Board Name:	Project Name:	
	Orion Main	Startracker Orion	
Sheet Path:	File Name:	Designer:	Date:
/Project Architecture/Power/	power.kicad_sch	Bertalan Nagy	
Sheet:	Reviewer:	Size:	Revision:
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Comments:	Company:	Variant:	Git Hash: e462fed	
	Board Name: Orion Main		Project Name: Startracker Orion	
Sheet Title:	File Name: power.kicad_sch	Designer: Bertalan Nagy	Date:	Revision:
Sheet Path: /Project Architecture/Power/		Reviewer:	Size: A4	Sheet: 5 of 7

[6] Connectors



	Comments:	Company:		Variant:	Git Hash: e462fed
	Sheet Title:	File Name: conn.kicad_sch		Project Name: Startracker Orion	
	Sheet Path: /Project Architecture/Connectors/	Designer: Bertalan Nagy		Date:	Revision:
		Reviewer:		Size: A4	Sheet: 6 of 7

